

Behavioural response of wild rodents to the calls of an owl: a comparative study

Colin A. Hendrie¹, Scott M. Weiss^{1,2} and David Eilam^{3*}

¹Department of Psychology, University of Leeds, Leeds LS2 9JT, U.K.

²Present address: Cerebrus Ltd, Silwood Park, Ascot, Berkshire, SL5 7PN, U.K.

³Department of Zoology, Tel-Aviv University, Ramat-Aviv 69 978, Israel

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Abstract

Effective anti-predator defence mechanisms depend on efficient recognition of possible predators. However, nocturnal aerial predators, such as owls, present very few stimuli to enable their detection by potential prey. The most prominent of these are their territorial calls which are believed to provide sufficient stimuli to activate anti-predator defence mechanisms in species heavily predated upon by owls. To test this hypothesis, the behaviour of voles (*Microtus socialis*), dormice (*Eliomys melanurus*), jerboas (*Jaculus jaculus*), and spiny mice (*Acomys cahirinus*) was analysed during and following exposure to playback of tape recorded calls of the tawny owl (*Strix aluco*). In response, voles crouched, and retreated to a small burrow present in the test arena; dormice also reduced the distances they travelled in the arena, and spent an increased amount of time around the burrow; jerboas crouched and froze during the playback of the owl call but not thereafter; spiny mice, however, displayed enigmatic behaviour in apparently ignoring the owl calls. The results, which demonstrate an induction of defensive behaviours to recorded owl calls, support the results of others who have examined the effects of exposing wild rodents to live owls.

Key words: predation, anti-predator defence, anxiety, stress, stereotypy, behavioural resource depression

INTRODUCTION

Initial defence depends on the ability of a prey species to recognize its predators (Endler, 1986) and the hunting strategy of the predator affects the defensive strategy of the prey. For example, weasels demonstrate an active-search predation pattern, employing sight and then smell to locate their prey, running in irregular patterns searching likely parts of the microenvironment for rodents (Powell, 1978). In contrast, tawny owls (*Strix aluco*) usually wait on a fixed perch and then swoop down on nearby prey. Barn owls (*Tyto alba*) often hunt on the wing (Martin, 1990).

In response to such pressures, prey species may be expected to develop predator-dependent defensive strategies appropriate to the hunting strategy of a particular predator. When exposed to a variety of predation pressure they may also be expected to develop differentiating mechanisms. For example, killing behaviour in weasels is stimulated by movement (Powell, 1978), whereas movement inhibits an owl from striking, as owls use the last sound location to predict the position

of the prey (Knudsen, 1980). In consequence, small rodents, such as woodmice, freeze or leap when exposed to stoats (Erlinge, Bergsten & Kristansson, 1974) and usually bolt for a hole when exposed to other predators (King, 1985). Such predator recognition and differentiation mechanisms can also be unequivocally demonstrated in a variety of species. Vervet monkeys (*Cercopithecus aethiops*) emit alarm calls which vary according to the presence of a ground-based or an aerial predator, which in turn elicit appropriate defensive behaviour in other members of the troop (Seyfarth & Cheney, 1980, 1986). It has also been shown that domestic chickens (*Gallus gallus*) produce alarm calls that may be differentiated along the same lines (Collias, 1987; Evans, Macedonia & Marler, 1993; Gyger, Marler & Pickert, 1987). Finally, wild rodents have been shown to display differential foraging and activity patterns following exposure to an owl or a snake (Kotler, Blaustein & Brown, 1992; Abramsky *et al.*, 1996).

Having initiated the final attack sequence, the success rate of many predators rarely falls below 90% (Curio, 1976), allowing strictly limited opportunities for learning. Thus, the working hypothesis of the present study is that in order to survive, rodents must have either innate or learned acquired ability to identify and react appropriately to the presence of a predator.

*All correspondence to: Dr David Eilam, Department of Zoology, Tel-Aviv University, Ramat-Aviv 69 978, Israel.
E-mail: eilam@ccsg.tau.ac.il

The main indication of owl activity in a given vicinity are their calls before flying to their hunting perches. Therefore, it is reasonable to assume that rodents learn from the calls about the presence of an owl. Indeed, in support of this hypothesis, studies in the field and in semi-natural environments have revealed that gerbils (*Gerbillus allenbyi* and *Gerbillus pyramidum*) reduce their visits to food sources when exposed to owls or their calls (Kotler *et al.*, 1992; Abramsky *et al.*, 1996). Laboratory studies have also shown that exposure of laboratory mice to owl calls induces activation of endogenous opioid-mediated analgesia systems (Hendrie, 1991) and inhibits feeding behaviours (Hendrie & Neill, 1991). Importantly, such responses have been shown to be relatively specific to owl calls, particularly tawny owl (*Strix aluco*) calls (see Hendrie & Weiss, 1994 for review). Tawny owls are primary small-rodent predators taking up to 8–10 woodland mice per night, and eventually killing more than 30% of a given population within a two-month period (King, 1985). However, although alterations in foraging patterns of the prey have been examined following their exposure to these raptors, there has as yet been no detailed examination of the specific behavioural anti-predator defence strategies adopted by small rodents.

The present study was designed to investigate the effects of exposure to tawny owl calls on the behaviour of wild rodent species in a laboratory set-up. Since the selected rodent species vary in their locomotor capabilities, feeding strategies, and habitat selection, a comparative study of their responses to owl calls may reveal the generality or the species specificity of this anti-predator behaviour.

METHODS

Animals and housing

Voles (*Microtus socialis*): 37–50 g (10 cm long, excluding tail) burrow-dwelling rodents that feed on seeds and green vegetation. Their eyes and external ears are small. The limbs and tail are short (2–3 cm). Because of their high fecundity, voles are a major agricultural pest and are a primary food source for owls.

Dormice (*Eliomys melanurus*): 48–54 g (13 cm long) rodents which inhabit rocky arid zones, nest between rocks/stones and feed on seeds, insects and snails. Eyes and external ears are large and the 11 cm long tail is black and bushy.

Jerboas (*Jaculus jaculus*): 52–61 g (11 cm long) bipedal rodents. Their hindlegs are 4–6 times longer than the forelegs. Their eyes and external ears are very large and their tail (18 cm long) features a black and white hairy brush at the distal end. Jerboas forage over large areas and feed mainly on seeds and green vegetation.

Spiny mice (*Acomys cahirinus*): 38–44 g (11 cm long) agile rodents that are common in rocky mountains, where they feed mainly on insects and snails, and nest

between rocks and stones. The dorsal fur of the spiny mouse is, as the name suggests, spiny and their eyes and external ears are of moderate size. The tail (10 cm long) is bare. (All descriptions are based on Mendelssohn & Yom-Tov, 1987.)

All animals were captured in baited live traps, except for the jerboas which were net-caught. Voles were obtained from the Lod region of central Israel. Jerboas, spiny mice, and dormice were trapped near the Ramon region of the Negev desert. After trapping, animals were transported to the Meir Segal's Research Zoo, Tel-Aviv University, Israel, where they were housed in pairs (male and female) in steel cages (80 × 50 × 50 cm), under natural light/dark cycle, and fed on seeds and diced vegetables daily. Jerboa pairs were housed in enclosures (200 × 400 × 120 cm) rather than cages. Various objects (pots, wooden boxes, etc.) were placed within each cage to provide shelter. Observations were made of 27 voles, 7 dormice, 7 jerboas, and 16 spiny mice, all trapped in the wild 1–4 months before testing. Since animals were kept in pairs of male and female, it was simple to recognize each individually by gender and cage number.

Apparatus

Individual rodents were exposed to tape-recorded owl calls in a 30 × 40 × 30 cm box which contained an 8 × 20 × 6 cm chamber attached below the front of the box with a 3 × 3 cm entrance, to provide a false burrow. During testing, this false burrow was available as a shelter for all species except the jerboa, which in preliminary observations ignored the burrow, just stumbling by its opening. The entire apparatus was illuminated under red light (2 × 60 W). Sound was transmitted into the arena by means of a small loud-speaker (2.5 inch, 8 Ω, 0.2 W max) placed on the rear wall, 25 cm above floor level, and connected to a cassette recorder. The front wall of the box and the burrow consisted of transparent Plexiglas, to allow the movements of the animals to be recorded on videotape. The apparatus was placed in an empty room in front of a video camera (Sony CCD-v700E). Sound levels of the tape-recorded owl calls were standardized to about 60–70 dB.

Procedure

Animals were brought to the laboratory 1 h before testing, to habituate them to the testing environment. This was not, however, a habituation to the testing apparatus, which occurred only in the pre-stimulus period of the test session (see below). All testing took place under dim red light during the early-mid phase (17:00–21:00) of the light/dark cycle, when these rodents display a peak activity (Mendelssohn & Yom-Tov, 1987). If tested after these hours the animals simply remained in the burrow throughout. Sand from the

home cage of each individual animal was placed in the burrow to facilitate acclimation. At the start of testing each animal was gently placed into the burrow of the test apparatus, and exposed to the sound stimuli. Behaviour was recorded on videotape for the next 10 min. The test apparatus was then cleared of faecal boli and urine, and wiped with a cloth in readiness for the next animal. For studies in which the number of animals allowed the inclusion of more than one sound stimulus in the experimental design, the animals were allocated to each of the conditions in randomized counter-balanced order and tested once only.

The 10-min testing procedure was divided into 3 consecutive periods:

1. Pre-stimulus period: 3 min playback of a human voice reading monotonically;
2. Stimulus period: 2 min playback of tawny owl (*Strix aluco*) calls (obtained from the British Sound Archive);
3. Post-stimulus period: 5 min silence.

These parameters were chosen to provide (a) 3 min during which animals were exposed to a neutral stimulus and had an opportunity to habituate to the test apparatus; (b) 2 min of the territorial calls of owls, to provide the test animal with as much information concerning the presence of an owl as would be available to it in the wild; and (c) 5 min of silence, to simulate the hunting pattern of the tawny owl which, following a period of territorial calling, flies to a hunting perch where it waits in complete silence to pounce on its prey.

Two types of stimuli were used as controls when animal numbers permitted: firstly the 2-min owl-call stimulus was replaced with an additional 2 min of human voice, secondly the 2-min owl-call stimulus was replaced with a backward-play of the owl-call (a reversed WAV file of the owl call). Accordingly, groups were labelled as *Forward* (test group exposed to normal owl call), *Backward* (groups exposed to reversed owl call), and *Human* (group exposed to human voice).

Behavioural analysis

Digital time-code (Telcom research, Burlington, Canada) was recorded on the videotapes together with the images of each animal, allowing accurate identification of each frame on screen. Two custom-designed computer programs (Hindsight, S. M. Weiss, unpublished; Motion, H. Szechtman & D. Eilam, unpublished) allow the analysis of the incidence and duration of various behavioural criteria and locomotor patterns. In the present study, these analyses include the following variables, calculated for each test period separately:

1. Total locomoted distance: The metric distance which the animals travelled. This was further divided into locomotion in the centre of the apparatus, and the counterpart of locomotion along the perimeter (the walls of the apparatus; further from burrow);

2. Burrow time: The cumulative time spent inside the burrow;
3. Home time: The cumulative time spent inside the burrow and/or the burrow opening;
4. Crouching: Except for the jerboa, this was scored as an immobile period, when no movement occurred. In the jerboa, crouching comprised immobility in a squat posture. In this posture, the jerboa was kneeling down and arching its back, which resulted in covering its white ventrum fur;
5. Rearing: Both fore paws off ground, and animal at least 2/3 erect. Rearing was terminated when all four paws returned to the ground. For the jerboas, bipedal rodents, rearing was scored when the animal showed a clear upward extension of its body and straightening of the legs and terminated upon resuming of the normal upright posture;
6. Sniffing: Animal was stationary (no forward progression), with the snout directed either at the floor or a wall, in contact or very near to contact), and displaying slight vertical or lateral head and/or torso scanning movements.

It should be noted that not all of these variables are presented in the Results. Rather we present for each species the variables which were significantly affected by the owl call, as compared to control stimuli.

Statistics

Since pre-stimulus, stimulus and post-stimulus period comprised different durations, data were first standardized per minute, and then analysed using single- or two-factor analysis of variance (ANOVA) with follow-up tests being performed using the appropriate error term from ANOVA. Where parametric assumptions were not met, analysis was performed by means of the non-parametric equivalents of ANOVA and appropriate follow-up tests.

RESULTS

Voles

Burrow time

Twenty-seven animals were randomly allocated to be exposed to 3 min of human voice followed by 2 min of either human voice, tawny owl call, or backward tawny-owl call ($n=9$ animals per group). These data failed to meet parametric assumptions for time spent in the burrow, and were therefore analysed using the Kruskal–Wallis test followed by the Mann–Whitney *U*-test. These analyses revealed a significant increase in burrow time in the post-stimulus period, after exposure to the forward owl call as compared to human voice or backward calls ($U(9,9) = 17$, $P < 0.05$; Fig. 1a).

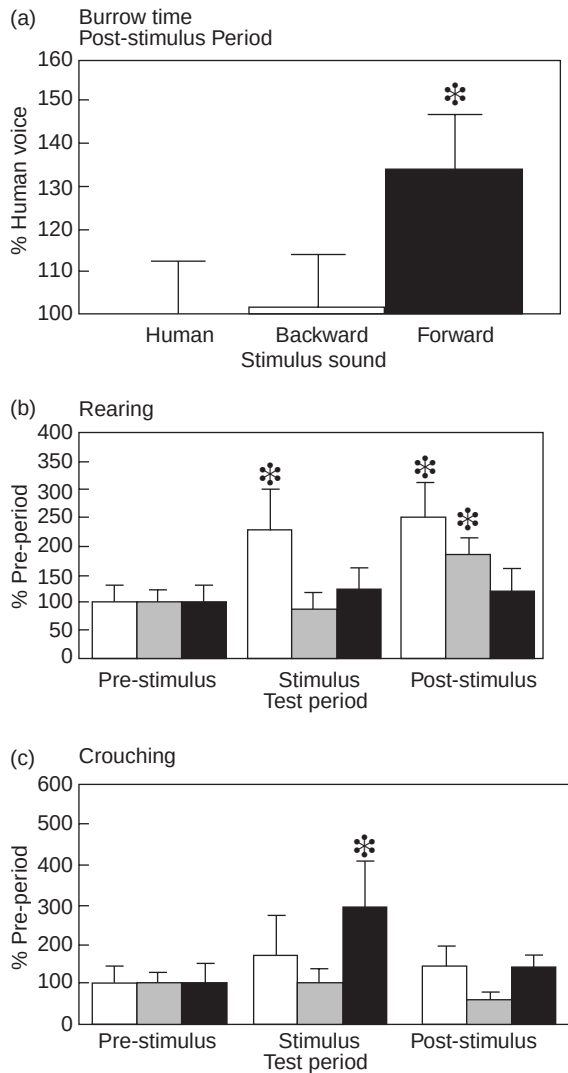


Fig. 1. Behavioural effects in the voles (*Microtus socialis*). Data are presented as means ($\pm$ SEM) of time spent in each activity. Data are expressed as a percentage of the time spent in each behaviour during exposure to the initial 3 min of a human voice. For each behaviour, data are described for the three test periods: pre-stimulus (first 3 min); stimulus (next 2 min), which was the forward owl in the test group, and backward call or a human voice in the control groups; and post-stimulus (last 5 min). $\square$, human-voice group; $\blacksquare$, the backward owl-call group; $\blacksquare$, the forward owl-call group. (a) Burrow time: a significant increase in forward owl-call group, as compared to each of the control groups. (b) Rearing: although during the pre-stimulus period rearing duration was similar in the three groups, during the stimulus period, the duration was shorter in both groups exposed to forward and backward owl calls, as compared to those exposed to human voice. In the post-stimulus period, the duration of rearing in the group exposed to forward call was significantly shorter compared to both control groups. (c) Crouching: crouching duration was significantly longer only during the playback of a forward owl call in the stimulus period. * (a) $P < 0.05$ from human voice control; (b) & (c) $P < 0.05$ from pre-stimulus period in the same playback condition.

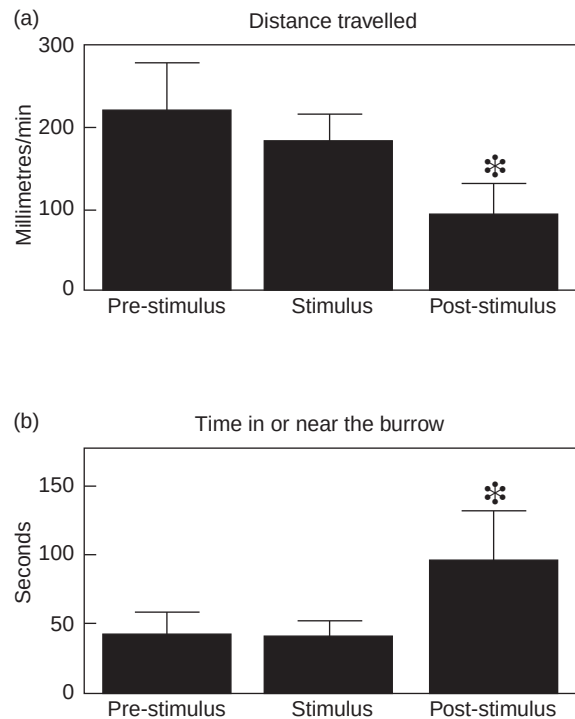


Fig. 2. Behavioural effects in the dormice (*Eliomys melanurus*). Data are presented for each period as means ($\pm$ SEM) of: (a) distance travelled (mm/min); and (b) time spent at the opening or inside the burrow. A significant decrease in the distance travelled and increase in the time near or in the burrow are evident in the post-stimulus period. * $P < 0.05$ from pre-stimulus period.

Rearing

Two-factor ANOVA (repeated measures on last factor) revealed significant effects of forward call ($F_{(4,48)} = 5.54$, $P < 0.01$), and follow-up tests using Dunnett's t indicated that the duration of rearing in the first 3 min was significantly less than that seen in the second 2 min ($t_d(48) = 3.54$, $P < 0.05$) and the final 5 min ($t_d(48) = 4.08$, $P < 0.05$). In animals exposed to the backward call, the duration of rearing seen in the final 5 min was significantly higher ($t_d(48) = 3.4$, $P < 0.05$). These results suggest that under these conditions rearing duration increases across the test session in all groups except those exposed to the forward tawny owl call (Fig. 1b).

Crouching

Animals exposed to forward call showed a significant increase in crouching during the stimulus period ($t_d(48) = 2.98$, $P < 0.01$; Fig. 1c).

Together these data indicate that across time, animals crouch during exposure to the owl call, do not increase rearing as in other stimulus conditions and spend a greater time in the burrow during the post-call period. Examples of trajectories of locomotion are provided in Fig. 4.

Dormice

Analysis of locomotor patterns revealed a significant decrease in distance travelled during the stimulus period (Friedman ANOVA_(2,7) = 9.85, $P < 0.01$; Fig. 2a) which follow-up tests using Wilcoxon matched pairs revealed to be due to a significant difference in distance travelled during the pre-stimulus and post-stimulus periods only ($Z = 2.36$, $P < 0.05$). Analysis of time spent in the burrow or around the burrow entrance (home time) also revealed significant effects of the stimulus period (Friedman ANOVA_(2,7) = 8.3, $P < 0.05$; Fig. 2b) which was, in this case, found to be due to a significant increase in home time from pre-stimulus to post-stimulus period ($Z = 2.201$, $P < 0.05$). Examples of trajectories of locomotion are provided in Fig. 4.

Jerboa

ANOVA revealed significant effects of stimulus on the duration of locomotion ($F_{(2,12)} = 42.17$, $P < 0.0001$; Fig. 3a) which follow-up tests revealed to be due to significant reductions, as compared to pre-stimulus measures, during the stimulus ($t_d(12) = 11.32$, $P < 0.001$) and post-stimulus periods ($t_d(12) = 11.16$, $P < 0.001$). There were also significant reductions, as compared to pre-stimulus, in distance travelled in the centre of the apparatus during the stimulus ($t_d(11) = 3.02$, $P < 0.05$) and post-stimulus periods ($t_d(11) = 2.33$, $P < 0.05$; Fig. 3b). Further reductions in the stimulus period were found on measures of rearing ($t_d(12) = 2.37$, $P < 0.05$) and sniffing the substrate ($t_d(12) = 2.69$, $P < 0.01$) (Fig. 3c, d). By contrast, ANOVA revealed a significant effect of stimulus on crouching ($F_{(2,12)} = 4.64$, $P < 0.05$), which was found to be due to a significant increase in this measure during the stimulus period only ($t_d(12) = 4.64$, $P < 0.05$; Fig. 3e).

These data indicate significant reductions in locomotion, rearing, and sniffing the substrate during the stimulus period, contrasted by a significant increase in crouching. During the post-stimulus period, there was recovery to pre-stimulus levels in all these measures except locomotion, for which there was also a change in spatial distribution in the centre of the apparatus. Examples of trajectories of locomotion are provided in Fig. 4.

Spiny mice

ANOVA or non-parametric analyses failed to show significant effects on any measure examined, which included time spent in the burrow or at the opening (home time), locomotor distance, locomotion in the centre of the apparatus, rearing, crouching and frequency of locomotion (data not shown). These data indicate that exposure to the owl call failed to induce any measurable/observable changes in the behaviour of spiny mice in the stimulus or post-stimulus periods.

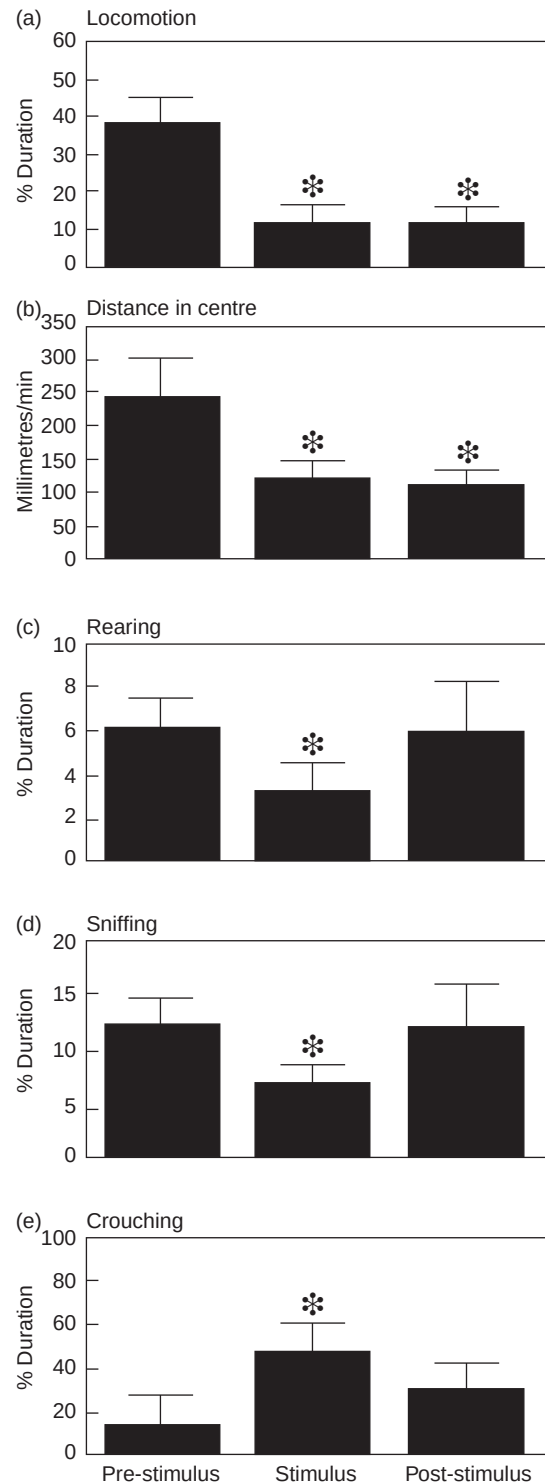


Fig. 3. Behavioural effects in jerboas (*Jaculus jaculus*). A significant decrease in the total duration of locomotion (a) and in the distance travelled in the centre of the arena (b) were obvious in the stimulus and post-stimulus periods. A significant decrease in the duration of rearing (c) and sniffing (d) during stimulus was followed in the post-stimulus period by a recovery to the pre-stimulus level. Crouching duration (e) significantly increased in the stimulus period. * $P < 0.05$ from pre-stimulus period.

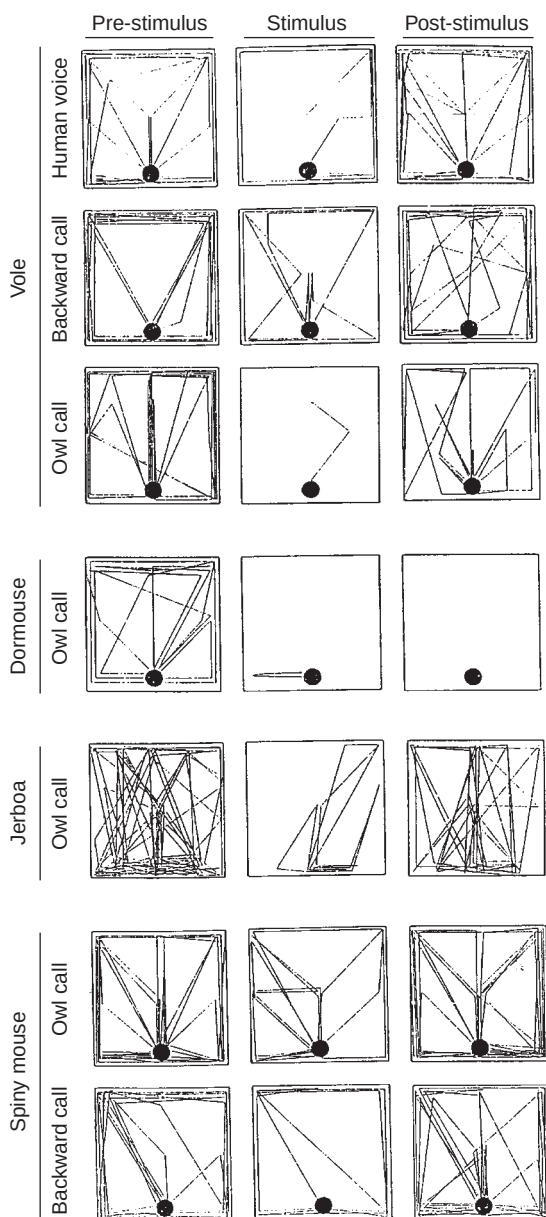


Fig. 4. Actual trajectories of locomotion in representative animals under each treatment. For each animal, the trajectories of locomotion are described for the entire period (without standardizing): pre-stimulus, 3 min; stimulus, 2 min; post-stimulus, 5 min. Columns are different periods, and rows are individual animals. ●, burrow opening. Voles: the decrease in movement during the stimulus period extends to the post-stimulus period and this vole displays a total decrease in the travelled distance especially along the perimeter of the arena, compared to the voles of the other groups and to its own behaviour in the pre-stimulus period. Dormice: went out of the burrow once, otherwise spending the entire stimulus and post-stimulus inside the burrow. Jerboa: a decrease in movement is evident in the stimulus period and in the 5 min of post-stimulus as compared to the 3 min of the pre-stimulus. Spiny mice: no difference is apparent between the animal exposed to a forward owl call and the animal exposed to a backward call. In each animal, the difference between the three periods seems a mere reflection of the different duration of each period.

Examples of trajectories of locomotion are provided in Fig. 4.

DISCUSSION

Our results demonstrate that voles, dormice, and jerboa react to the playback of recorded calls of the tawny owl, whereas spiny mice, under our test conditions, do not. The data also reveal certain common features in the reaction to owl calls amongst a variety of species occupying different niches. In general, all species studied, with the exception of spiny mice, show reduced levels of activity and increased attention towards the strategically defensible burrow location during exposure to the call and immediately after.

Specifically, voles showed increased crouching during exposure to the owl call and spent more time in the burrow during the post-stimulus period. Their levels of rearing were also comparatively reduced. Similarly, dormice demonstrated a decrease in distance travelled during the stimulus and post-stimulus periods and spent a greater percentage of their time in or near the burrow during these phases of the test. Although in view of their unusual morphology, the burrow was not provided in studies employing jerboa, they none the less showed a similar pattern to voles and dormice; i.e. during the stimulus period, there were significant reductions in locomotion, rearing, and sniffing the substrate. These effects were correlated with a significant increase in crouching. Levels of locomotion also remained low during the post-stimulus period and there was an additional reduction in activity in the centre of the apparatus.

Broadly speaking, our findings may be viewed as generalized reactions to novel stimuli. The failure of reversed owl calls to induce the same behavioural changes induced by forward owl calls questions, but does not preclude this interpretation. In voles, dormice, and jerboa it was apparent that changes in behaviour were related to the different stimulus periods. This specificity of response to forward owl calls is further supported by observations in laboratory mice (Hendrie & Weiss, 1994). Moreover, behavioural reactions observed after stimulus presentation had ended must have been because of changes induced by the stimulus presented. Importantly, decreased locomotion continued to be observed in voles, dormice, and jerboas following stimulus presentation, although changes in rearing, sniffing the substrate and crouching could also return to pre-stimulus levels in this period. These data suggest that the patterns of behaviour observed during and after exposure to the owl call are related to the nature of the stimuli, rather than novelty *per se*.

To address the functional significance of these responses, we must consider the hunting pattern of the owl. Tawny owls, having made their territorial calls, fly to a different location to hunt. Their preferred method is to 'perch and pounce', and therefore the only indication that a potential prey species has of the presence of a

tawny owl is the calls it makes before hunting. Consequently, the time immediately following hearing a call is when there is the greatest probability of attack. When illumination levels are low, tawny owls use the termination of the last locatable sound source to predict the position of their prey; otherwise they use sight, which is about 100 times more sensitive to light than diurnally active birds and overlaps with the sensitivity of cats (Mikkola, 1983).

The most adaptive strategies of prey animals upon detecting the presence of an owl are to remove themselves from the immediate location, shelter in strategically defensible places or to alter their patterns of movement (Blanchard & Blanchard, 1989). They must either constantly move to avoid their location being predicted, or minimize their movements to avoid being detected. The reduction in locomotor activity observed in voles, dormice, and jerboa beginning in the stimulus period and continuing into the post-stimulus phase, thus appears to serve to reduce the probability of detection. These findings contrast markedly with observations in laboratory mice, in which protean behaviour is commonly expressed (Hendrie, 1991). Furthermore, voles, as burrow dwellers, show a marked increase in the time spent in this strategically defensible location. Dormice, which live in rocky areas, with many small hiding places, spend more of their time in or near the burrow during the stimulus and post-stimulus periods. Jerboas reduce their levels of locomotion and avoid the relatively exposed central area of the test apparatus. Importantly, they also show increased crouching behaviour, which serves not only to minimize movement but also to hide their relatively light-coloured ventral surface.

In the context of field studies involving the examination of predator–prey interactions, our findings are in good agreement with responses seen in gerbils following exposure to barn owls (*Tyto alba*), in which alterations in foraging time and patterns have been observed. In these studies, the presence of owls decreased the frequency of foraging trips to exposed sites and increased the number of visits to protected sites under bushes (Kotler *et al.*, 1992). This process is known as behavioural resource depression and has clear parallels with our data, in which exposure to owl calls induces a reduction in locomotor activity and increased attention to strategically defensible locations. Were such reactions to occur under natural circumstances, this would have a clear effect on foraging activity. Interestingly the lack of response of spiny mice to owl calls may be addressed within the same framework. Voles and jerboa are seed-eating animals and hence must, as a consequence of normal foraging behaviour, expose themselves to potential predatory attack. In contrast, spiny mice primarily eat snails, and may thus be afforded some protection from predatory attack by the microenvironment in which they forage. As such, there is little reason for them to interrupt their normal activities upon hearing an owl call. Except for the anomalous behaviour of spiny mice, it may be predicted that the intensity of the reaction to an owl call, which may be related to

behavioural resource depression, is related to the need of each animal to forage in open areas.

The lack of response in spiny mice is puzzling considering the effect seen in the other species. This lack of response is even more striking for three additional reasons. First, dormice also feed on snails and inhabit the same niches as spiny mice, yet dormice did react significantly to owl calls while spiny mice did not. Second, spiny mice are predated in the field by owls (T. Dayan, pers. comm.). Caged tawny owls were also offered two tanks (70 × 70 × 70 cm): one with 10 spiny mice, and the other with 10 voles. Surprisingly, the owls ate eight of the spiny mice but only two of the voles, indicating that owls do feed on spiny mice when they are available. Therefore spiny mice would be expected to react to owl calls. Third, we have recently studied whether owl calls induce an increase in stress hormones in voles and spiny mice (Eilam *et al.*, in prep.). While we have found a significant increase in stress hormones in voles which were exposed to owl calls, there was no such an increase in spiny mice. Altogether, our evidence suggests that further studies are required to uncover the reasons for the anomalous lack of response in spiny mice.

In the context of behavioural resource depression, although field studies demonstrate that such a phenomenon exists, the present data indicate ways in which this may be effected. In the field, close observations of the behaviour of small rodents are difficult, resulting in few opportunities for detailed behavioural analyses. Our findings indicate that reduced foraging behaviour following exposure to an owl may be secondary to increased expression of defensive behaviours, rather than a direct influence on feeding behaviour *per se*. Such a suggestion is difficult to evaluate experimentally, but none the less it is clear that increases in defensive behaviour must be correlated with reductions in other activities, including feeding. Thus, our laboratory work complements field studies by providing a direct description of the changes in the individual patterns of behaviour induced by owl calls, while field studies provide the broader ecological implications which are involved in the mediation of these behavioural changes. Together, both approaches provide a more comprehensive view of the multi-faceted interaction between tawny owls and their prey.

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